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Sub ject : DSA

Unit 2, S11, SLO2 - Doubly Linked List – Deletion at Middle

Algorithm to delete a node after a given node NUM

Step 1: IF START = NULL

Write UNDERFLOW

Go to Step 9

[END OF IF]

Step 2: SET PTR = _____

Step 3: Repeat Step 4 while PTR -> DATA != _____

Step 4: SET PTR = PTR -> NEXT

[END OF LOOP]

Step 5: SET TEMP = _____

Step 6: SET PTR -> NEXT = _____

Step 7: SET _____ = PTR

Step 8: FREE _____

Step 9: EXIT

Answer: Step 2: SET PTR = START

Step 3: while PTR -> DATA != NUM

Step 4: SET PTR = PTR -> NEXT

Step 5: SET TEMP = PTR -> NEXT

Step 6: SET PTR -> NEXT = TEMP -> NEXT

Step 7: SET TEMP -> NEXT -> PREV = PTR

Step 8: FREE TEMP

Step 9: EXIT

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[END OF IF]

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[END OF LOOP]

Step 5: SET TEMP = _____

Step 6: SET _____ = PTR

Step 7: SET _____ = TEMP -> PREV

Step 8: FREE _____

Step 9: EXIT

Answer: Step 2: SET PTR = START

Step 3: while PTR -> DATA != NUM

Step 4: SET PTR = PTR -> NEXT

Step 5: SET TEMP = PTR -> PREV

Step 6: SET TEMP -> PREV -> NEXT = PTR

Step 7: SET PTR -> PREV = TEMP -> PREV

Step 8: FREE TEMP

Step 9: EXIT